

# The Race between Man and Machine

Tasks, wages, and what a formal proof can establish

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<https://github.com/johnbarraza/ai-06-restrepo>

# Road map for 20 minutes

- ① The aggregate task economy and equilibrium threshold
- ② Automation versus new tasks: opposite reallocations
- ③ The wage condition, its analytical and numerical check
- ④ Lean translation: checked theorem, proof step, and limits
- ⑤ Hand verification and verdict

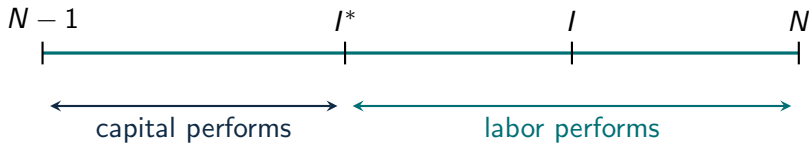
# Question and unit of analysis

## Question

Can automation raise output yet lower the wage? Can new tasks reverse the fall in labor's share?

Unlike the earlier weekly papers, the unit here is the **aggregate economy**: a continuum of production tasks and competitive factor markets.

# The task space



## Two frontiers

$I$  is technological feasibility;  $I^*$  is actual capital use. They need not coincide.

## From tasks to aggregate output

$$Y = \tilde{B} \left( \int_{N-1}^N y(i)^{(\sigma-1)/\sigma} di \right)^{\sigma/(\sigma-1)}.$$

- ▶ The task interval always has measure one.
- ▶  $\sigma > 0$  governs substitution across tasks; the  $\sigma = 1$  case is understood by its limit.
- ▶ Automation changes *who performs* tasks; new task creation shifts the interval to higher  $i$ .

The resulting aggregate production function has technology-dependent capital and labor weights (Proposition 1).

# The agents' problems

**Competitive task producer** For  $i \leq I$ ,  
compare factor costs

$$\min \left\{ R, \frac{W}{\gamma(i)} \right\}.$$

For  $i > I$ , only labor is feasible. Intermediate inputs also enter full unit cost.

**Representative household** Choose  $(C, L)$   
with  $C = WL + RK$  to maximize the paper's  
 $u(C, L)$ . At an interior choice:

$$\nu'(L) = \frac{W}{C}.$$

## Comparative advantage implies a threshold

$$\frac{W}{R} = \gamma(\tilde{I}), \quad I^* = \min\{I, \tilde{I}\}.$$

- ▶  $\gamma(i)$  strictly increases: labor has an advantage in more complex tasks.
- ▶ Below  $I^*$ : capital is cheaper; above  $I^*$ : labor is used.
- ▶ If  $I < \tilde{I}$ , technology constrains automation.
- ▶ If  $\tilde{I} < I$ , technology is slack; more feasible automation changes nothing locally.

# Conditions behind the static results

- ①  $\gamma(i)$  strictly increasing (Assumption 1).
- ②  $\eta \rightarrow 0$  or  $\zeta = 1$  (Assumption 2): homothetic factor demand.
- ③  $K < \bar{K}$  (Assumption 3): new tasks are used.

$\sigma > 0$ ,  $\hat{\sigma} > 0$ ,  $\varepsilon_L > 0$ , and the household's convexity and regularity conditions also matter. Proposition 1 supplies a unique static equilibrium under these conditions.

# Automation and reinstatement

## Automation: $I \uparrow$

Capital takes over some existing labor tasks. In the constrained regime,  $W/R$ , labor share, and employment fall.

## New tasks: $N \uparrow$

New complex tasks require labor and replace low-index tasks.  $W/R$ , labor share, employment, and the wage rise.

These are Proposition 2–3 static comparative statics, not unconditional predictions for a dynamic economy.

## Where the wage equation comes from

$$(B9) \quad s_L d \ln W + (1 - s_L) d \ln R = d \ln Y|_{K,L},$$

$$(B10), dN = 0 : \quad d \ln W - d \ln R = -\frac{\Lambda_I}{\hat{\sigma} + \varepsilon_L} dl.$$

Let  $x = d \ln W$ ,  $z = d \ln R$ , and  $p = d \ln Y|_{K,L}$ . Since  $z = x - (x - z)$ , (B9) becomes

$$x = p + (1 - s_L)(x - z).$$

Substituting (B10) gives the Proposition 3 wage decomposition. This two-equation algebra is the natural Lean target.

## The wage trap: two forces

$$\underbrace{\frac{d \ln W}{dl}}_{\text{wage response}} = \underbrace{P_I}_{\text{productivity gain}} - \underbrace{\frac{(1 - s_L)\Lambda_I}{\hat{\sigma} + \varepsilon_L}}_{\text{displacement}}.$$

Exact local condition,  $I^* = I < \tilde{I}$

The wage rises iff

$$P_I > \frac{(1 - s_L)\Lambda_I}{\hat{\sigma} + \varepsilon_L}.$$

Equality gives no local change; a reversed inequality gives a fall.

## A different sign is fixed

$$\frac{d \ln(W/R)}{dl} = -\frac{\Lambda_l}{\hat{\sigma} + \varepsilon_L} < 0.$$

The labor share and employment move with  $\omega = W/(RK)$  and therefore fall in the constrained case.

### Interpretation

A falling labor share does not prove that the *level* of the real wage falls. Productivity and displacement compete in the wage equation.

## Analytical and computational check

| Illustrative case       | $P_I$ | displacement | $d \ln W / dI$ |
|-------------------------|-------|--------------|----------------|
| Large productivity gain | .400  | .133         | +.267          |
| Small productivity gain | .050  | .133         | -.083          |
| Equality                | .133  | .133         | 0              |

Same  $s_L = .6$ ,  $\Lambda_I = .5$ ,  $\hat{\sigma} = 1$ ,  $\varepsilon_L = .5$  in every row. Reproduce with python `analysis/wage_condition.py`. These inputs illustrate the sign identity; they are not solved equilibria.

# Where I did not trust the AI

## Claim to test

“Automation necessarily reduces wages and labor’s share.”

- ▶ Labor share falls only when the automation frontier binds; at a slack frontier, a small increase in  $I$  has no effect.
- ▶ The wage can rise when the productivity gain dominates displacement.
- ▶ At  $I^* = I = \tilde{I}$ , use one-sided derivatives.

## Hand check: the decisive inequality

Real photograph of the student's handwritten derivation pending. See `hand/README.md`.

$d \ln W / dI = P_I - D_I$ , so a wage increase requires  $P_I > D_I$ .

## Lean 1/5: what entered the checker

### Pinned source

NBER Working Paper 22252, June 2017, 87 PDF pages. SHA-256 starts 441d0120.

- ▶ AppliedModelingLib generated the complete AR18RaceManMachine paper folder.
- ▶ The Lean target is Appendix B's algebra from (B9) and (B10) to the wage formula.
- ▶ The code treats the differential values as real variables and takes (B9), (B10) as explicit hypotheses.

This is a checked **support lemma**, not a proof that the economy satisfies those equations.

## Lean 2/5: equation $\rightarrow$ statement $\rightarrow$ proof

$$\underbrace{s_L d \ln W + (1 - s_L) d \ln R = p}_{(B9)}, \quad \underbrace{d \ln W - d \ln R = q}_{(B10)} \implies d \ln W = p + (1 - s_L)q.$$

```
theorem wage_identity_from_B9_B10
  (s dW dR productivity relativePrice : Real)
  (hB9 : s*dW + (1-s)*dR = productivity)
  (hB10 : dW-dR = relativePrice) :
  dW = productivity + (1-s)*relativePrice := by
  linear_combination hB9 + (1-s)*hB10
```

$s$  encodes labor's net-output share;  $dW$ ,  $dR$  encode the two log changes;  $p$  is productivity;  $q$  is the relative-price change. The slide spells Lean's  $\mathbb{R}$  notation as `Real`. Lean verifies the algebra *given* both premises.

## Lean 3/5: why the proof line works

Write (B9) and (B10) as zero-valued residuals:

$$\begin{aligned}A &= s \, dW + (1 - s) \, dR - p, \\ B &= dW - dR - q.\end{aligned}$$

The Lean proof uses  $A + (1 - s)B = 0$ . The  $dR$  terms cancel and  $s \, dW + (1 - s)dW = dW$ , leaving

$$dW - p - (1 - s)q = 0.$$

`linear_combination` checks that weighted sum symbolically; it does not infer (B9) or (B10) from the task model.

## Lean 4/5: the exact wage condition

Set  $q = -\Lambda_I/(\hat{\sigma} + \varepsilon_L)$  for the binding-frontier case. The checked specialization gives

$$d \ln W = P_I - \frac{(1 - s_L)\Lambda_I}{\hat{\sigma} + \varepsilon_L}.$$

*Excerpt below: binders and repeated arguments omitted for legibility.*

```
theorem automation_wage_rises_iff ... :  
  0 < dW <-> (1-s)*lambdaI/denominator  
    < productivity := by  
  rw [automation_wage_change ...]  
  constructor <;> intro h <;> linarith
```

The proposition is conditional on the encoded (B9) and (B10) identities. Its variables range over  $\mathbb{R}$ ; positivity of  $\hat{\sigma} + \varepsilon_L$  is needed to derive (B10)'s sign from the economic primitives, not for this algebraic equivalence.

## Lean 5/5: results and open boundary

### Machine results

```
lake build +AR18RaceManMachine: passed.  
paper_contribution.py check AR18RaceManMachine --fast: passed.  
Three support theorems use only propext, Classical.choice, and Quot.sound.
```

### What remains open

The full continuum equilibrium, its threshold comparative statics, and the derivation of (B9) and (B10) from the paper's model. No named result is claimed as fully proved. Source-to-Lean review and final closeout were not run.

## What the checks can and cannot conclude

- ▶ The computational script verifies arithmetic on the stated wage identity.
- ▶ Lean's compiler can verify a formal theorem only under its encoded assumptions and domain.
- ▶ Matching the NBER claim requires a separate source-to-Lean comparison.
- ▶ The full continuum equilibrium and all its comparative statics should be counted only if formally represented and proved.

# Conclusion

## Economic result

Automation reallocates tasks away from labor. With a binding frontier, labor share falls, while wages depend on the balance between productivity and displacement. New tasks reinstate labor.

## Methodological result

The threshold, regime conditions, and sign inequality must all be explicit in any Lean translation or oral explanation.

# Sources

Acemoglu, D., and P. Restrepo (2018). “The Race between Man and Machine: Implications of Technology for Growth, Factor Shares, and Employment.” *AER* 108(6), 1488–1542.

Formalization source: NBER Working Paper 22252, revised June 2017, 87 pages. In particular Section 2, Propositions 1–3 and Appendix B.

Course requirement: <https://github.com/alexanderquispe/AI-Econ-Modeling/issues/5>